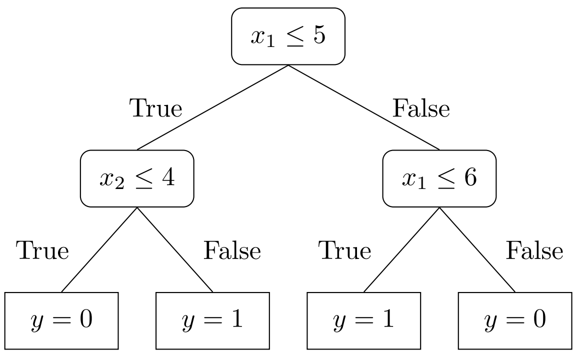
CSC311 Introduction to Machine Learning   Week 2 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

W2-T1. Given the decision tree below, what would the prediction be for a data point $\mathbf{x} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$?



W2-T2. Draw the decision boundary for the above tree. Clearly label the prediction in each region.

W2-T3. Suppose that we have the following joint distribution:

$$
\begin{aligned}
P(X = 0, Y = 0) &= 0.1 \\
P(X = 0, Y = 1) &= 0.25 \\
P(X = 1, Y = 0) &= 0.25 \\
P(X = 1, Y = 1) &= 0.4
\end{aligned}
$$

(a) What is the entropy $H(X)$?

(b) What is the entropy $H(Y)$?

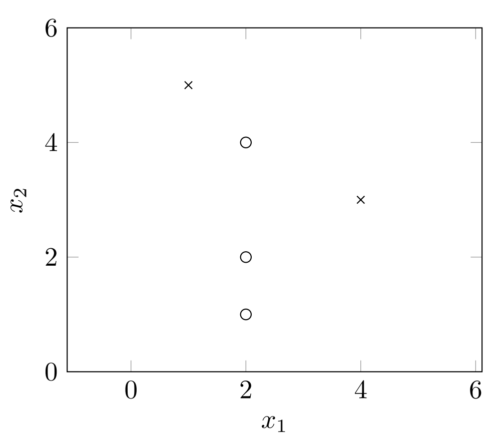
(c) What is the conditional entropy $H(Y|X = 1)$?

(d) What is the conditional entropy $H(Y|X = 0)$?

(e) Compute the expected conditional entropy $H(Y|X)$.

(f) Is $H(Y|X)$ the same value as $H(X|Y)$?

W2-T4. We wish to build a decision tree classifier on the below data set. Compute the information gain of the split $x_1 \geq 3$. Is the split $x_1 \geq 3$ better than the split $x_2 \geq 4$? Why or why not?



W2-T5. Prove the following properties of the entropy H and expected conditional entropy $H(Y|X)$ of a discrete random variable.

(a) The entropy H is always non-negative.

(b) $H(X,Y) = H(X|Y) + H(Y) = H(Y|X) + H(X)$

(c) If X and Y independent, then $H(Y|X) = H(Y)$

(d) By knowing Y makes our knowledge of Y certain: $H(Y|Y) = 0$

W2-T6. A decision tree is a **universal function approximator**. What does this mean?

W2-T7. Show, with an example, that a split can increase the information gain, even if the split does not decrease misclassification rate.

Validation

W2-V1. What are the advantages/disadvantages of a tree that is deep (vs shallow)? Your analysis should consider underfitting/overfitting, computational cost of training/making predictions, and the interpretability of the model.

W2-V2. Which of these statements are true? Briefly justify your answer.

(a) On the same training set, a deeper tree generally has a higher training accuracy than a shallow one.

(b) If the decision tree's greedy algorithm selects at a node a split along a feature x_j, then any of its descendent nodes will never be a split along that same feature x_j.

(c) A decision tree split will always put equal number of training data points in its two branches.

(d) Shallow trees are more likely to underfit the data.

W2-V3. **This is an important one.** You've been given a dataset of your company's employees performance evaluations and asked to design a system to predict performance for potential employees. The performance measure is a rating on a scale of $\{1,2,3,4\}$ with 1 being the worst performance and 4 being the best. The dataset contains the following features for each employee: education level (bachelors or masters) and whether they have prior work experience (yes/no).

education	experience	performance
bachelors	no	1
bachelors	yes	3
masters	yes	4
bachelors	no	1
masters	no	1
bachelors	no	2
masters	yes	4
bachelors	yes	4
masters	no	2
bachelors	yes	3
masters	no	2
bachelors	no	3

(a) Using information gain as your feature selection mechanism, construct a decision tree that predicts a potential new employee's performance based on the input attributes of education and experience. Continue splitting until no more attributes are available, or until all data agrees on the target attribute (performance). Use the majority class at each leaf as your point estimate (prediction). In case of a tie, choose the lower performance number. Show your work (i.e. information gain calculations) and draw a graph of your final decision tree.

(b) What prediction would your decision tree give for the following 3 test vectors?

education	experience	performance
masters	yes	
bachelors	no	
masters	no	

Generalization

W2-G1. In what situation is a k-nearest neighbour model more suitable than a decision tree model? This is a broad question. You might think in terms of data set size, time/space complexity for learning, time/space complexity for inference, etc.

W2-G2. A fair coin is flipped until the first head appears. Let X denote the number of flips required.

(a) What is $P(X = 4)$?

(b) What is $P(X = n)$ for any n?

(c) What is $H(X)$, the entropy of X? You might find it useful to recall that $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ and $\sum_{n=0}^{\infty} nr^n = \frac{r}{(1-r)^2}$ for $0 < r < 1$.

(d) What is $\mathbb{E}[X]$, the expected value of X?

W2-G3. In this question, we will show that $\mathrm{IG}(Y|X)$ is non-negative. That is, by knowing X, we can only decrease uncertainty about Y: $H(Y|X) \leq H(Y)$

(a) Show that $IG(Y|X) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$.

(b) Given two probability distributions p and q, their *Kullback-Leibler divergence* is defined as follows:

$$KL(p|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$$

This is a measure of how *different* the distributions p and q are. In particular, it can be shown that $KL(p|q) \geq 0$ and $KL(p|q) = 0$ if and only if $p = q$ for any distributiosn p and q. Using these properties of the KL divergence, showo that $IG(Y|X) \geq 0$. (Hint: Given $p(x,y)$, let $q(x,y) = p(x)q(y)$.)

(c) Conclude that $H(Y) \geq H(Y|X)$.